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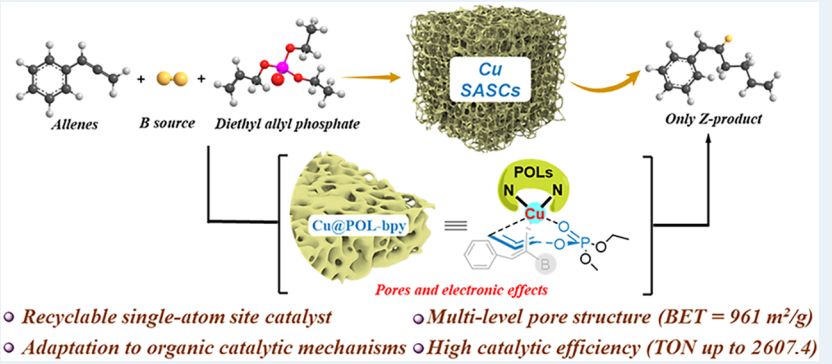
Air-Stable Copper(I) Single-Atom Site Catalysts for Highly Selective Borylative Allyl−Allyl Cross-Coupling

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ABSTRACT: Borylative allyl−allyl coupling reactions of allenes are an efficient strategy for constructing 1,5-diene skeletons. Nevertheless, this reaction poses challenges in achieving precise chemical selectivity and stereochemical control. We designed an air-stable copper(I) single-atom site catalyst (SASC), Cu@POL-bpy, with high regioselectivity for borylative allyl−allyl coupling of aryl-allenes. This breakthrough resulted from thorough investigations into structure−performance relationships, highlighting the importance of the hierarchical porosity and metal−support interactions of the catalyst. The catalytic system had a surface area of 961 m^2/g and attained a turnover number (TON) of 2607.4, representing almost a 150-fold improvement over previous catalytic systems. This study highlights the promise of SASCs in organic synthesis and provides valuable insights and tools for advancing synthetic chemistry.

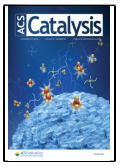
KEYWORDS: *single-atom site catalysts, allyl−allyl cross-coupling, organic ligand porous polymer, allenes, high selective addition*

INTRODUCTION

Despite advances in synthesis, efficiently constructing complex alkenes with precise stereocontrol remains challenging.[1−14] In particular, the 1,5-diene framework constitutes the basic structural motif found in many biologically active natural products and drug compounds (Figure 1A).[15−17] Although the reaction between allyl nucleophiles and electrophilic reagents provides a direct pathway for preparing 1,5-diene, the use of stoichiometric organometallic reagents in traditional methods is extremely limited. In contrast, copper (Cu)-catalyzed borylative coupling offers a more advantageous strategy: Cu−Bpin addition to an unsaturated hydrocarbon generates a nucleophilic organocopper intermediate in situ, which undergoes electrophilic trapping, allowing the synergistic formation of both C−C and C−B bonds.[18−20] A central challenge of this methodology is precisely controlling regio- and stereoselectivity, particularly in the allylic borylation subtype. This process involves the coupling of in situ-formed Cu-Bpin intermediates with allylic electrophiles. Due to the potential reactivity at either the α- or γ-position, various regioisomers are theoretically accessible, a situation further complicated by stereochemical heterogeneity (Figure 1B).

Hoveyda and Tsuji independently reported the Cu-catalyzed coupling reaction between allyl phosphonates and allyl boranes. Within this Cu-catalyzed system, the reaction proceeds with favorable regioselectivity and stereoselectivity to yield α,β-addition coupling products. However, this approach is primarily applicable to alkyl-allenes, with limited success in achieving highly selective transformation of aryl-allenes.[21,22] This is because aryl-allenes possess higher π-electron density than alkyl-allenes. Nevertheless, this higher electron density is mainly concentrated in the C═C bond region of the allenes rather than being delocalized across the

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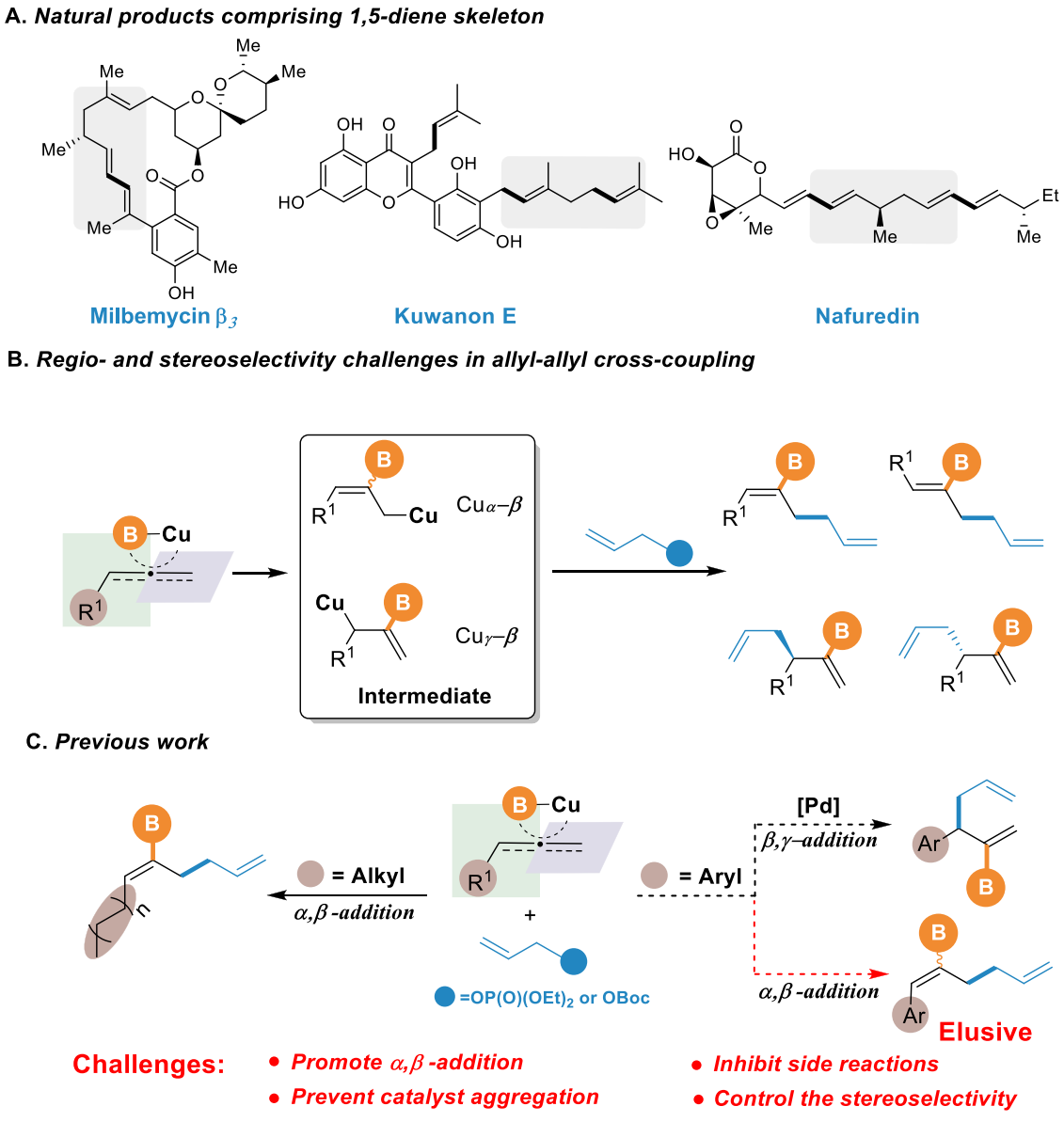


Figure 1. (A–C) Metal-catalyzed borylative allyl–allyl cross-coupling of allenes.

entire molecular framework, creating substantial obstacles for attaining high regio- and stereoselectivity. Although several Cu/Pd synergistic catalytic systems have been developed to achieve high selectivity for aryl-allenes, they predominantly afford β,γ-addition coupling products (Figure 1C).[23,24] Consequently, achieving a highly selective α,β-addition coupling reaction with aryl-allenes remains a considerable challenge.

In recent years, Cu single-atom site catalysts (SASCs) have shown great promise in terms of increasing chemical selectivity and catalytic efficiency because of their distinct coordination environment and exceptional physicochemical properties.[25−49] Presently, considerable advancements have been made in the development of SASC systems based on stable valence-state metals such as Cu^{2+}. However, the design and synthesis of catalytic systems incorporating nonstable valence-state metals, such as Cu^{+}, continue to pose substantial technical challenges. These challenges mainly stem from the need to ensure the structural integrity and stability of metal active sites. To overcome this challenge, we designed a bipyridine (bpy)-derived porous organic ligand polymer (POL) support via strategic vinyl functionalization of the aromatic framework. Using a metal–ligand exchange approach, we precisely anchored Cu atoms onto the POL support, obtaining a novel air-stable Cu@POL-bpy catalyst that maintained intact ligand functionality.[50,51] Detailed structure–activity studies of aryl-allene borylative coupling revealed two critical features of the catalyst. First, the hierarchical porous architecture facilitates reactant diffusion/mass transport while accommodating sterically diverse substrates.[52,53] Second, strong metal–support interactions regulate the electronic structure of the catalytic centers while preventing metal aggregation (Figure 2A).[54−57] These synergistic effects endow Cu@POL-bpy with outstanding catalytic performance (Figure 2B). This study expands the range of applications for SASCs in organic catalysis and offers valuable theoretical and experimental guidance for designing and preparing the next generation of high-performance catalysts.

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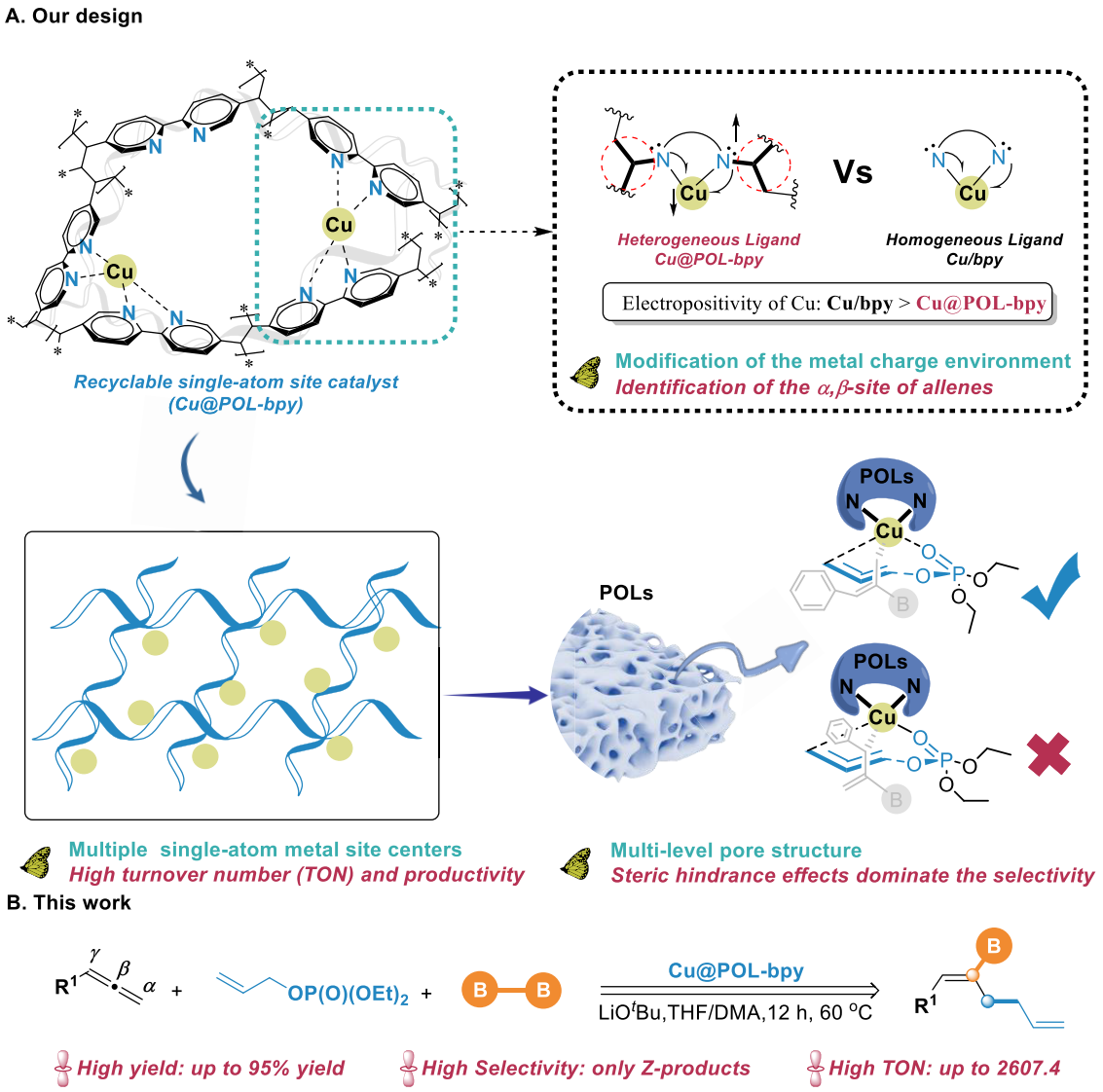


Figure 2. (A, B) Design concept of single-atom site catalysts and this work.

METHODS AND MATERIALS

Conditional Screening and Materials Research. Our initial optimization experiments employed a homogeneous catalytic system, with systematic screening of key parameters: ligands (Table S1, entries 1−10), Cu salts (Table S1, entries 11−15), solvent systems (Table S2, entries 1−7), bases (Table S2, entries 8−12), and temperature conditions (Table S2, entries 14−16). Control experiments confirmed the essential role of the base because no product formation occurred in its absence (Table S2, entry 13). Using systematic optimization, the homogeneous catalytic system employing CuCl with the bpy ligand in a tetrahydrofuran/*N*,*N*-dimethylacetamide solvent mixture (60 °C, 12 h) afforded the target product in 52% yield (Table S1, entry 3). The moderate yield and selectivity underscored the need for advanced catalyst design to overcome the inherent constraints of homogeneous systems. We developed POL-bpy using rational molecular design. Employing a metal−ligand exchange strategy, we achieved precise anchoring of single-atom Cu sites, which effectively catalyzed the challenging aryl-allenes α,β-addition coupling reaction. The resulting SASCs demonstrated high catalytic performance, delivering a 90% yield with excellent stereoselectivity (Table S3).

Detailed characterization revealed that the exceptional performance originates from the hierarchical porosity and strong metal−support interactions. Although catalytic efficacy was established, advanced characterization of Cu@POL-bpy was required to elucidate the structure−activity relationships at the atomic level. The thermal gravimetric (TG) curve revealed outstanding thermal stability of the catalyst, as evidenced by the TG curve showing maintained structural integrity under high-temperature conditions (up to 400 °C) (Figure 3A). Comprehensive porosity characterization was conducted via nitrogen adsorption−desorption curve measurements, providing a quantitative assessment of both specific surface area and pore size distribution (Figure 3B). Notably, the Cu@POL-bpy material demonstrated excellent textural properties, featuring a surface area of 961 m^2/g and a broad pore size distribution spanning from mesoporous to macroporous regimes with pore diameters between 2.3 and 30.0 nm (Figure 3C). The large specific surface area enabled optimal reactant adsorption capacity, serving as a crucial determinant for the improved catalytic performance and reaction yields. Complementary

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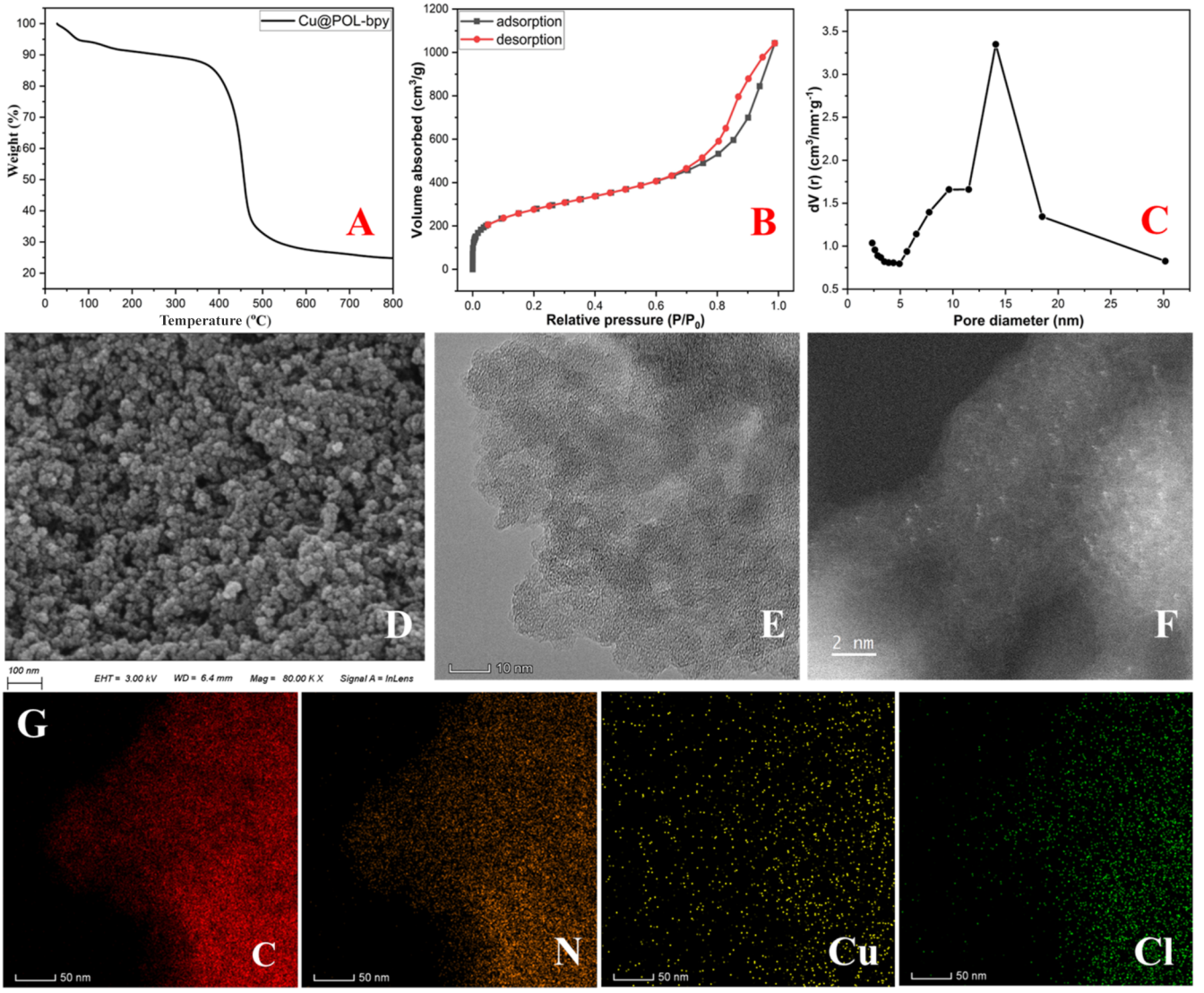


Figure 3. Structural characterization of Cu@POL-bpy. (A) TG of the Cu@POL-bpy. (B) Nitrogen adsorption−desorption curve of Cu@POL-bpy. (C) Pore size distribution of Cu@POL-bpy. (D) SEM of Cu@POL-bpy. (E) TEM of Cu@POL-bpy. (F) AC HAADF−STEM of Cu@POL-bpy. (G) EDS−mapping of Cu@POL-bpy.

scanning electron microscopy (SEM) characterization was performed to elucidate the hierarchical pore architecture of Cu@POL-bpy. SEM clearly demonstrated the well-defined multiscale porosity spanning from nano- to microscale dimensions (Figures 3D and S1).

Transmission electron microscopy (TEM) of the Cu@POL-bpy material indicated that no clusters developed during the catalyst preparation and synthesis process (Figure 3E). The Aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (AC HAADF−STEM) images showed that the Cu atoms in Cu@POL-bpy were uniformly dispersed and densely packed, with no aggregation observed (Figure 3F). Energy dispersive spectrometry mapping (EDS−mapping) revealed that Cu elements were uniformly dispersed in Cu@POL-bpy (Figure 3G).

To further investigate the microstructure and coordination mode of the catalyst, we conducted X-ray photoelectron spectroscopy (XPS). According to the XPS spectra, the binding energy of N 1s decreased from 398.12 to 398.06 eV (Figure S2). This decrease in binding energy indicated that N formed coordination bonds with Cu, adsorbing electrons from Cu atoms around the N atoms. Additionally, in the Cu 2p XPS spectrum of Cu@POL-bpy, the peaks at 931.5 eV corresponded to Cu $2p_{3/2}$, whereas those at 950.8 eV represented characteristic peaks of Cu $2p_{1/2}$ (Figure 4A). Notably, XPS analysis revealed distinct binding energy shifts for Cu species with different valence states loaded on the same support materials. These findings offer preliminary evidence indicating that the Cu species in Cu@POL-bpy predominantly existed as the Cu^{+} species (Figure S3). X-ray absorption near-edge structure (XANES) spectral analysis showed that the absorption edge position of the Cu@POL-bpy catalyst was closer to that of copper(I) oxide (Figure 4B). The Fourier-transformed extended X-ray absorption fine structure (FT−EXAFS) spectra further revealed the coordination environment of Cu (Figure 4C and Table S4). Based on the fitting results, it can be tentatively concluded that the Cu atoms in the catalyst were mainly coordinated by $Cu-N_4$.[41,58] Wavelet transform analysis clearly excluded the presence of Cu−Cu bonds (Figure 4D,E), confirming that the Cu was dispersed in

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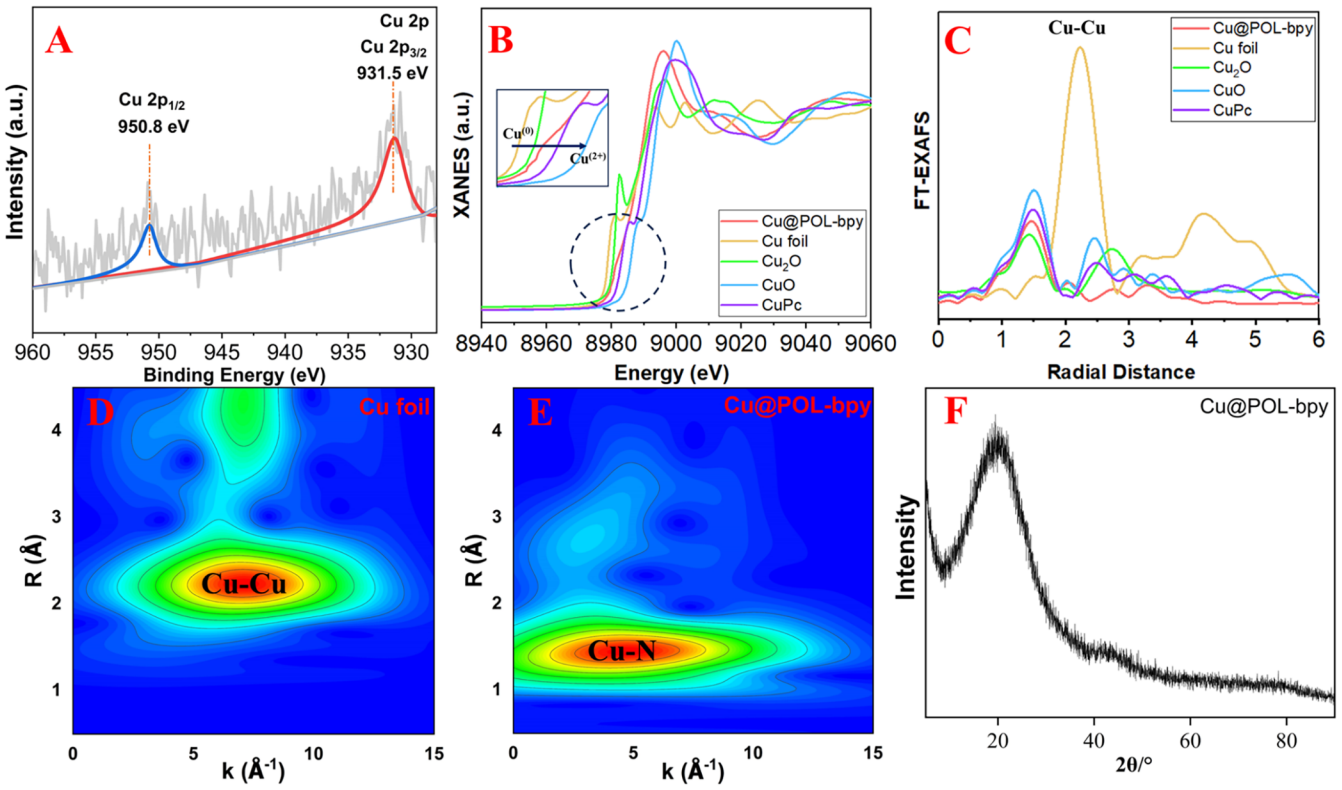


Figure 4. Structural characterization of Cu@POL-bpy. (A) XPS of Cu 2p on the Cu@POL-bpy. (B) Cu *K*-edge XANES of Cu@POL-bpy and reference samples. (C) *R*-space FT−EXAFS of Cu@POL-bpy. (D, E) The wavelet transform of the Cu foil and Cu@POL-bpy. (F) XRD of Cu@POL-bpy.

single-atom form. No characteristic diffraction peaks were observed in the X-ray diffraction (XRD) pattern of Cu@POL-bpy, indicating an amorphous structure and further confirming the absence of Cu nanoparticles in the catalyst (Figure 4F). A quantitative analysis employing inductively coupled plasma atomic emission spectrometry (ICP−AES) revealed that the Cu content of Cu@POL-bpy was 0.0858 wt %, confirming the production of a low-loading SASC. These findings consistently confirmed the highly dispersed distribution of Cu atoms within the catalyst material and further substantiated the reliability of the ICP−AES experimental data. Notably, based on the ICP−AES measurements, Cu@POL-bpy exhibited highly favorable catalytic efficiency, achieving a TON as high as 2607.4, which was ∼150 times greater than the previous record (Table S5).

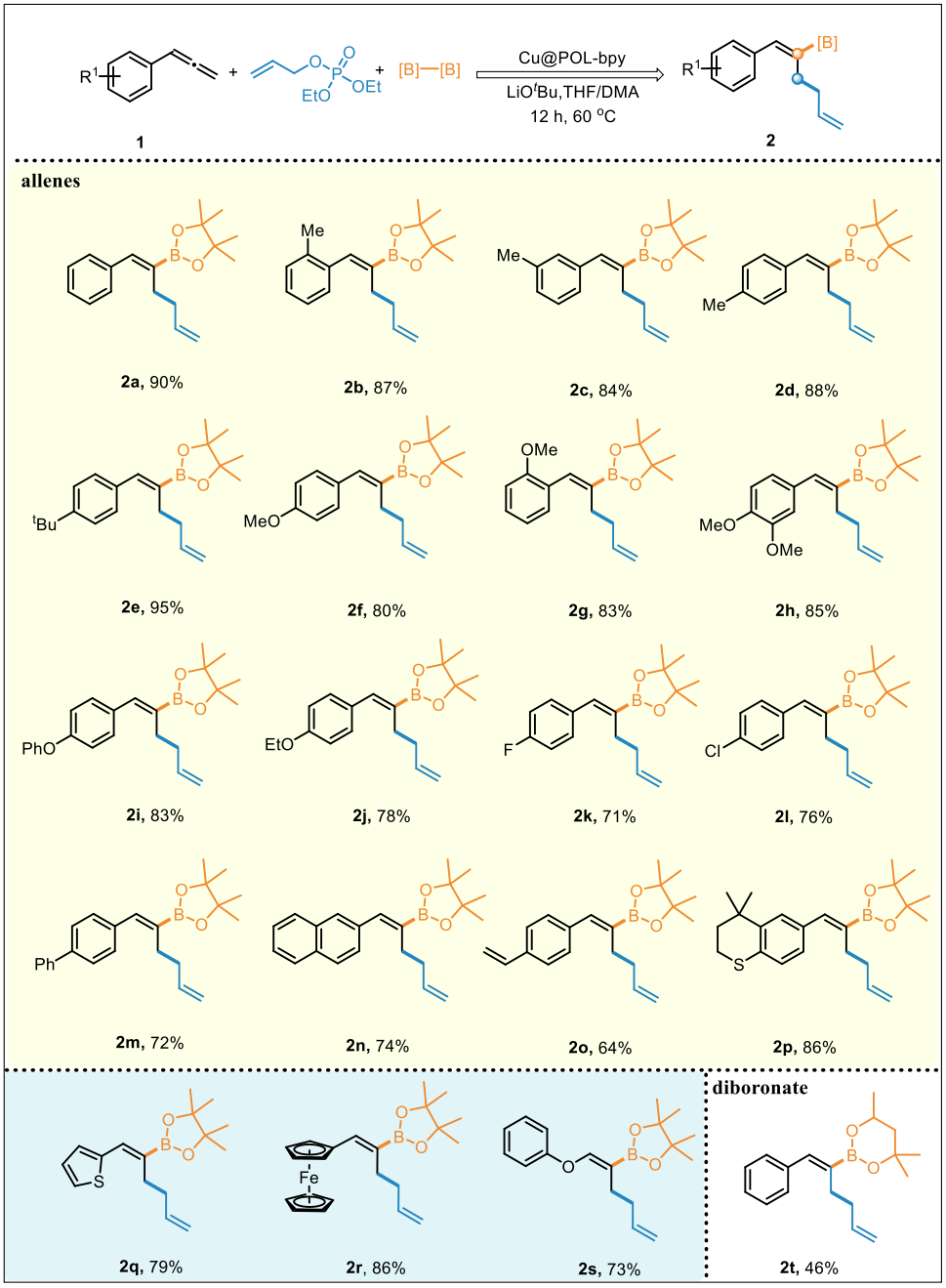
To evaluate the reusability of the catalyst, we selected substrate **1a** as a representative example. The heterogeneous catalyst was recovered by sequentially washing and filtering it after each catalytic run. The catalyst maintained consistent activity over five consecutive reaction cycles, demonstrating excellent recyclability (Figure S4). Comprehensive characterization including SEM (Figure S5), TEM (Figure S6), AC HAADF−STEM (Figure S7), Nitrogen adsorption−desorption analysis (Figure S8) and EDS−mapping (Figure S9) verified preservation of the porous architecture and the absence of metal aggregation. Additionally, we conducted a comparative experiment to examine the leaching effect. After reacting for 6 h, the heterogeneous catalyst was separated from the reaction mixture via filtration, and the reaction assay was continued for various durations. Notably, the reaction yield remained constant in the catalyst-free filtrate (Figure S10). These findings indicated that the coordination interaction between Cu and the carrier in the catalyst remained stable throughout the catalytic process.

Substrate Scope. Although the developed SASC demonstrated high catalytic efficiency, required a more systematic exploration of its substrate compatibility with diverse allenes. A thorough substrate scope investigation was therefore performed to assess catalytic versatility. The catalyst effectively transformed various allenes, including those with electron-donating (Scheme 1, **2b**−**2j** and **2p**) or electron -withdrawing (**2k** and **2l**) groups, biphenyl/naphthalene units (**2m** and **2n**), and heterocycles (**2q** and **2r**), all affording products in high yields. Notably, the substituent position on the aromatic rings had minimal influence on product yield (**2b**−**2d**). Alkyl-allenes (**2s**) and different boron reagents (**2t**) were used as substrates to generate products with a single configuration. In particular, in cases where both allenes and alkenes were present in the substrate, the catalyst demonstrated high selectivity by preferentially recognizing and catalyzing allenes (**2o**). In contrast, 1,1-di- and 1,3-disubstituted allenes did not produce the desired products selectively because of electronic or spatial effects (Table S6).

Applications. To comprehensively assess the synthetic potential of this reaction system, we systematically examined the scalability of the reaction and the practical applications of the products. First, the catalytic system demonstrated exceptional scalability in gram-scale experiments, successfully synthesizing the target compound **2e** with a yield of 92% at a 3.0 mmol scale (Scheme 2A). This outcome clearly confirmed the stability of the catalyst under scaled-up conditions and its industrial application potential. Second, using the coupling reaction between boron-substituted 1,5-

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Scheme 1. Scope of Copper-Catalyzed Highly Selective Synthesis of Borylative Allyl−Allyl Cross-coupling of Allenes[a,b]



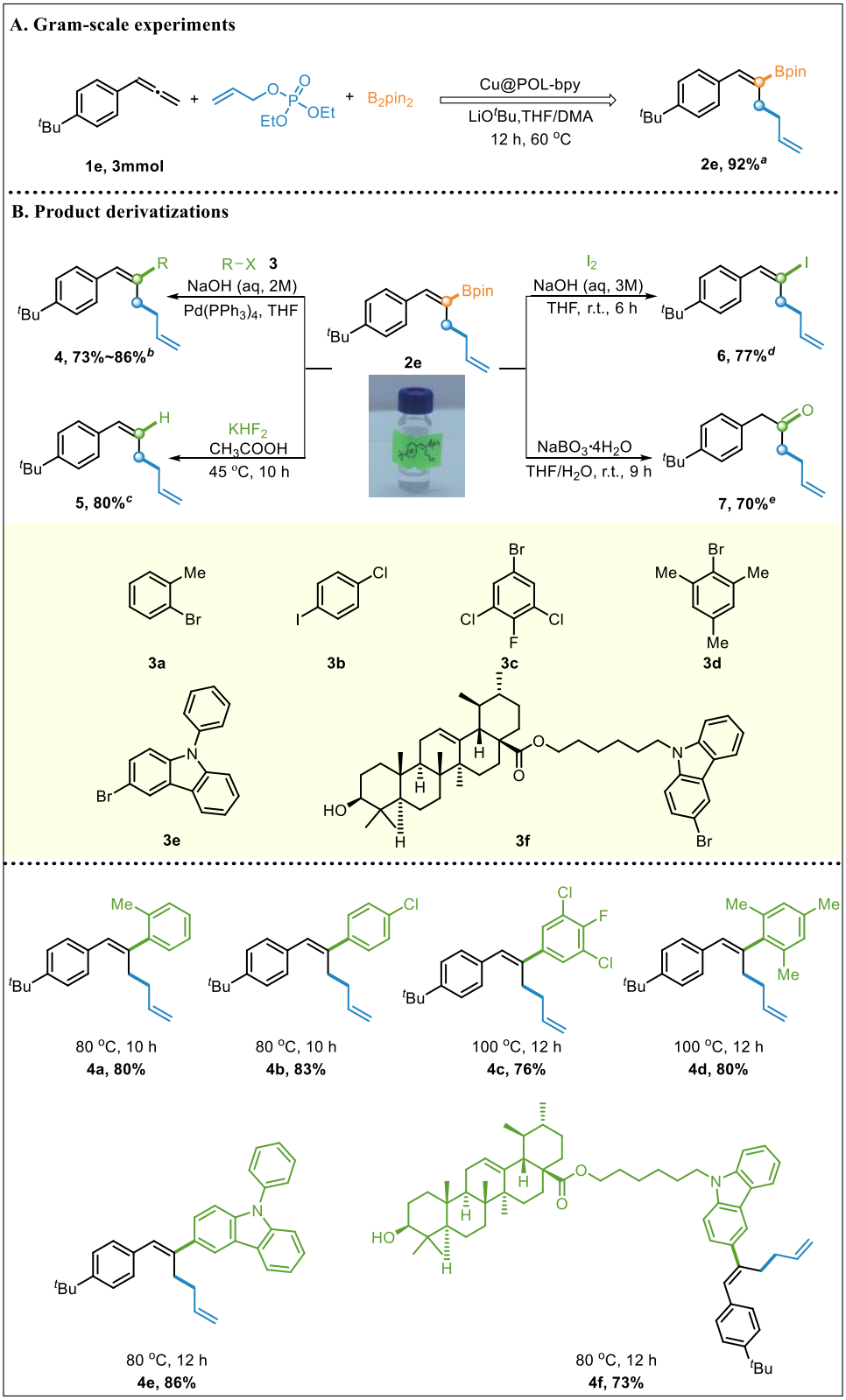
[a]Reaction conditions: **1a** (0.2 mmol), phosphate (0.5 mmol), B source (0.4 mmol), Cu@POL-bpy (5 mg), LiO^{t}Bu (0.5 mmol), THF/DMA = 9:1 (2 mL), 60 °C, 12 h. [b]Yields of isolated **2**.

dienes and aryl halides as an efficient method for constructing polysubstituted olefinic compounds, we performed coupling reactions of **2e** with various aryl halides and achieved high yields (Scheme 2B, **4a**−**4f**). Importantly, we succeeded in selectively modifying the anti-inflammatory drug ursolic acid by combining it with **2e**, yielding a novel ursolic acid derivative (**4f**). This approach not only extends the utility of **2e** in drug molecule modifications but also provides new opportunities for developing ursolic acid derivatives with greater bioactivity. Moreover, **2e** functions as a versatile synthetic building block for a variety of organic transformations. For instance, **2e** can be converted into specifically configured 1,5-dienes via deboronation reactions (**5**), transformed into iodides via iodination (**6**), or oxidized to ketones (**7**), which serve as key intermediates for subsequent cross-coupling reactions.

Mechanistic Investigation and Density Functional Theory (DFT) Calculations. We initially conducted a kinetic analysis of the reaction and observed that the reaction rate exhibited first-order dependence on the amount of catalyst employed (Figures S11 and S12). To further confirm the experimental results, a possible mechanism reaction was proposed and quantum chemistry was performed to optimize the theoretical steady-state structure of Cu@POL-bpy (Figure S13). The catalyst carrier possessed organic ligand-like properties, thereby facilitating catalytic behavior comparable to homogeneous catalysis. First, similar to homogeneous

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Scheme 2. (A, B) Exploring the Application Potential of the Reaction



[a]Reaction conditions: **1e** (3.0 mmol), diethyl allyl phosphate (0.75 mmol), B source (0.6 mmol), Cu@POL-bpy (50 mg), LiO^{t}Bu (0.75 mmol), THF/DMA = 9:1 (30 mL), 60 °C, 12 h. [b]**2e** (0.2 mmol), **3** (0.3 mmol), NaOH (2 M, 0.4 mmol), $Pd(PPh_3)_4$ (0.02 mmol), THF (2 mL). Isolated yields of **4** are shown. [c]**2e** (0.14 mmol), KHF_2 (0.42 mmol), HOAc (2 mL), 45 °C, 10 h. [d]**2e** (0.2 mmol), I_2 (0.4 mmol), NaOH (3 M, 0.6 mmol), THF (2 mL), r.t., 6 h. [e]**2e** (0.2 mmol), $NaBO_3{\cdot}4H_2O$ (1.2 mmol), THF (1 mL), H_2O (1 mL), r.t., 9 h.

catalysis, the base is a key component that initiates the reaction, and **intermediate A** is initially formed from Cu@POL-bpy and LiO^{t}Bu at the onset of the reaction (**step I**). This accounts for the lack of reactivity in the absence of a base.

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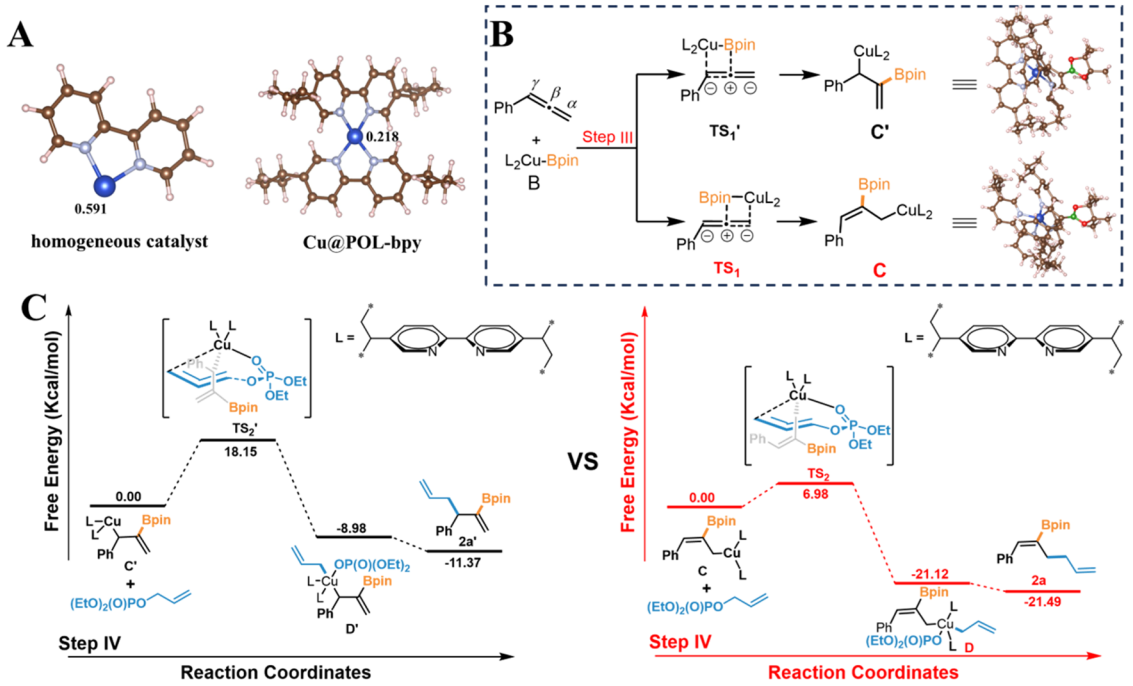


Figure 5. Electronic structure and reaction mechanism verification. (A) Charge density of metals in homogeneous catalyst and Cu@POL-bpy. (B) Comparison of differences between the two intermediates C and C'. (C) DFT calculations for the intermediate C(C') to **2a** (**2a'**) transformation process.

Then, **intermediate A** reacts with $B_2(pin)_2$ to obtain **intermediate B** (**step II**).[24] **Step III** is the selective addition reaction of **intermediate B** with the allene substrate to produce **intermediate C**. The DFT calculations showed that the charge density of the Cu active center decreased from 0.591 to 0.218 after introducing the polymer carrier. This electronic effect confirmed that the polymer carrier plays an electronic modulation role similar to that of organic ligands in the catalytic system, which increases the ability of the catalyst to recognize the ***α*-site** and ***γ*-site** of allenes (Figure 5A). Second, we compared the activation energy barriers of the two possible reaction pathways using transition state theory calculations. The results showed that when the catalyst binds to the ***α*-site** of allenes, the energy required is lower and the intermediate formed is more stable (Figures 5B and S14). **Step IV** involves the reaction of **intermediate C** with diethyl allyl phosphate via an S_N2'-type oxidative addition pathway to yield **intermediate D**.[22] Notably, the multistage pore structure of the catalyst exhibits a particular spatial site-barrier effect in this step, which becomes a key factor in controlling the selectivity. The calculated results showed that the formation of **intermediate C′** must overcome an energy barrier of 18.15 kcal/mol to form the transition state TS_2' with diethyl allyl phosphate, whereas the formation of **intermediate C** requires only 6.98 kcal/mol to form the transition state TS_2 (Figures 5C and S15). This energy difference causes the reaction to occur mainly via the TS_2 transition state and **intermediate D**, resulting in the target product **2a**. This theoretical calculation was in perfect agreement with the experimental data, confirming the key role of the unique spatial site resistance effect of the catalyst in controlling the reaction selectivity.

CONCLUSIONS

The rationally designed air-stable Cu@POL-bpy catalyst, featuring atomically dispersed Cu centers on bpy-based organic polymer supports, enables regio- and stereoselective borylative allyl−allyl *α*,*β*-addition coupling of aryl-allenes with exceptional efficiency (TON up to 2607.4). Beyond advancing the precise synthesis of multifunctional alkenes, this methodology provides a versatile platform for constructing structurally sophisticated molecules with tailored stereochemical profiles. The results confirm that the Cu@POL-bpy catalyst demonstrates excellent performance in selectivity modulation and exhibits considerable potential for application in organic synthesis. This study not only advances the utilization of SASCs in organic catalysis but also establishes a solid theoretical and experimental foundation for the future design and preparation of high-performance catalysts, representing a substantial milestone in synthetic chemistry.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c05647.

Experiment procedures and copies of ^1H and ^{13}C NMR spectra of products (PDF)

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Author Contributions

L.Z., H.-T.T. and Y.-M.P. conceived and designed the study. S.R., S.L. and Z.R. performed sample preparation, characterization and theoretical calculations. P.C., M.-R.W. and A.G. provided project guidance and supervision. S.R. and S.L. analyzed the data and wrote the manuscript. All authors discussed the results, reviewed and edited the manuscript, and approved the final version before submission.

Notes

The authors declare no competing financial interest.

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